[CH] — The Exact Markov Chain Solver

The Chain, MFPT, Committor, Spectrum, and the Four-Way Cross-Validation

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1 Why build the chain first

The master document's recommendation (C6) is that Project A is exactly solvable and should not be primarily a simulation project. The reason is worth stating precisely, because it changes what the student is doing all month.

For *any fixed policy* π, the joint process (x_{bin}, M) is a finite, time-homogeneous Markov chain. Every quantity the project asks for is then a linear-algebra problem on a sparse matrix of dimension $N = n\,2^b$:

Physical question	Linear-algebra object
Stationary density ρ_s, effective potential	leading left eigenvector of P^π
Mean first-passage time $\langle T_{\text{esc}} \rangle$	one sparse solve $(I - P^\pi_{\text{trans}})T = \mathbf{1}$
Committor, transition state, κ	second solve, *same* matrix
Full FPT distribution, non-exponentiality	spectrum of P^π_{trans}
True optimum for a memory class	optimise over π, objective exact

All of these are *exact*: no sampling error, no histogram fitting, no convergence diagnostics. Monte Carlo is then demoted from primary instrument to validation target, and Q-learning is demoted from discovery mechanism to a check on *learnability* against a known optimum. That is the correct scientific ordering.

2 1.5.1 — Discretization and construction of P^π

2.1 Position discretization, and a choice that makes the chain exact

Take a domain $[x_L, x_b]$ with n bins of width h, centres x_i. From (1) below, the one-step position update given action a is Gaussian:

$$x_{t+1} \mid (x_t = x, a_t = a) \;\sim\; \mathcal{N}\big(x + va\Delta t,\; 2D\Delta t\big), \tag{1}$$

so the position kernel is a difference of normal CDFs,

$$K_a(i,j) \;=\; \Phi\left(\frac{x_j + \frac{h}{2} - x_i - va\Delta t}{\sqrt{2D\Delta t}}\right) - \Phi\left(\frac{x_j - \frac{h}{2} - x_i - va\Delta t}{\sqrt{2D\Delta t}}\right). \tag{2}$$

Warning

The small-D trap. The headline prediction of §1.2 lives at $D \to 0$, where $\sqrt{2D\Delta t} \ll h$ and (2) degenerates: the deterministic displacement $va\Delta t$ must land on a grid point or the chain acquires a large, silent interpolation error. Naively refining n does *not* fix this — it makes it worse, since h shrinks while $va\Delta t$ does not.

Key result

Grid matching. Choose

$$h \;=\; \frac{v\Delta t}{K}, \qquad K \in \mathbb{N}.$$

The deterministic part of the step then moves the agent exactly K bins. In the important case $D = 0$, $K = 1$, the position is *always* exactly on a grid point, the kernel is $K_a(i,j) = \delta_{j,i+a}$, and the chain is not an approximation of the model at all — **it is the model, represented exactly**. The discretization error is identically zero.

This is the single most useful implementation decision in the module. It converts the MC-vs-chain comparison of §7 from "do these two approximations agree" into "does the simulator have a bug," which is a far sharper test.

2.2 The policy kernel: marginalising the observation

The agent acts on $\tilde{c} = c(x) + \sigma\eta$, not on x. For a memory state M the action distribution at bin i is

$$\pi(a \mid i, M) \;=\; \int \Pr\big(a \mid \tilde{c}, M\big)\, \mathcal{N}\big(\tilde{c};\, c(x_i),\, \sigma^2\big)\, d\tilde{c}. \tag{3}$$

This integral is where the sensing noise enters the chain, and it is done *once*, analytically or by quadrature, at build time. For the threshold 0-bit policy of §1.1.1 it is $\pi(+1 \mid i) = \Phi\big((c(x_i) - \theta)/\sigma\big)$ in closed form.

2.3 The memory kernel

The memory update $M_{t+1} = f(\tilde{c}_t, M_t)$ contributes a factor $q(M' \mid M, i, j, a)$. Crucially, for a temporal-comparison agent this factor depends on *both* endpoints i and j, because the quantity being judged is the concentration change across the step. It therefore does *not* factor out of the position kernel, and P^π is not a Kronecker product.

2.4 Assembly

Index states as $k = 2^b\, i + M$. Then P^π is a $2^b \times 2^b$ block matrix of $n \times n$ blocks,

$$\boxed{\big[P^\pi\big]_{(i,M)\to(j,M')} \;=\; \sum_a \pi(a \mid i, M)\; K_a(i, j)\; q\big(M' \mid M, i, j, a\big).} \tag{4}$$

Dimension $N = n2^b$; the number of nonzeros per row is (bandwidth of K_a) $\times\, 2^b \times |\mathcal{A}|$, so the matrix is sparse and banded. For $n = 2000$, $b = 1$ this is $N = 4000$ with a handful of nonzeros per row — a solve that takes milliseconds.

2.5 Worked case: the 1-bit temporal-comparison agent

This is the case that carries the project's central claim, so it is worth writing out completely. Recall from the audit in the companion document that the *independent-readings* design is the one whose closed forms are exact; it is also, as we now see, the one that admits a finite chain at all.

Take the state to be (i, a): current bin, and the action just executed. Then:

1. The agent arrived at bin i from bin $i' = i - a$ (at $D = 0$, $K = 1$). The *true* concentration change over that step is
$$\Delta c \;=\; c(x_i) - c(x_{i-a}), \tag{5}$$
computed exactly from the stored landscape — no gradient approximation, no expansion in Δt.

2. The observed change carries independent sensing errors on both readings, so $\Delta\tilde{c} \sim \mathcal{N}(\Delta c,\, 2\sigma^2)$ and
$$p_{\text{persist}}(i, a) \;=\; \Pr(\Delta\tilde{c} > 0) \;=\; \Phi\!\left(\frac{c(x_i) - c(x_{i-a})}{\sqrt{2}\,\sigma}\right). \tag{6}$$

3. The new action is $a' = a$ with probability p_{persist} and $a' = -a$ otherwise; the agent then moves to $j = i + a'$.

Key result

$$\left[P^{\pi}\right]_{(i,a)\to(i+a',\,a')} = \begin{cases} p_{\text{persist}}(i,a), & a' = a, \\ 1 - p_{\text{persist}}(i,a), & a' = -a, \end{cases} \qquad p_{\text{persist}} = \Phi\left(\frac{c(x_i) - c(x_{i-a})}{\sqrt{2}\sigma}\right). \tag{7}$$

Exactly $2n$ states, exactly two nonzeros per row, and *no* approximation whatsoever at $D = 0$.

Remark 1 (Why the single-sensor agent has no finite chain)**.** *If instead the agent stores the previous* reading $\tilde{c}_{t-1}$ *and differences it against* $\tilde{c}_t$ *(the design literally specified in the master document), then the memory is a* continuous *variable and the state space is* $x \times \mathbb{R}$*, not* $x \times \{0, 1\}$*. There is no finite chain, the exact solver does not exist, and every result in this document becomes unavailable. This is an independent and quite decisive argument for adopting the independent-readings design recommended in the companion audit: it is not merely analytically convenient, it is what makes the project exactly solvable. If the single-sensor variant is wanted as a comparison, it must be run by Monte Carlo only, or its stored reading discretized into additional bits (at which point it is a* $b > 1$ *agent, and should be described as one).*

2.6 Boundaries, and three sanity checks

Boundaries. Reflecting at x_L: clamp $j = \max(0, i + a')$. Absorbing at x_b: designate the bins with $x \geq x_b$ as an absorbing set $\mathcal{A}$ and leave their rows as identity (or simply delete them, as in §4). The absorbing set must be defined identically in the chain, the Monte Carlo, and the quadrature — see §7.3.

Checks to run before trusting anything.

(S1) **Row sums.** $\sum_j P_{kj} = 1$ for every transient k, to machine precision. This catches almost all indexing errors.

(S2) **Irreducibility.** Every transient state must reach $\mathcal{A}$. A disconnected region gives an infinite MFPT and a singular solve.

(S3) **Aperiodicity.** See the warning below — this one is easy to miss and it corrupts the spectral diagnostic rather than crashing.

Warning

Parity. With $D = 0$, $K = 1$ and $\mathcal{A} = \{\pm 1\}$, the position index changes by exactly ± 1 every step. The chain is therefore *bipartite*: $i + t$ has fixed parity, the chain has period 2, and P^{π} carries an eigenvalue near $-\lambda_1$ alongside λ_1. This is an artifact of the lattice, not physics. It does not affect the MFPT solve (§4), but it *will* wreck the non-exponentiality diagnostic of §6 if the second eigenvalue is taken by magnitude. Fixes, in order of preference: (i) allow a rest action $a = 0$; (ii) use $D > 0$ with $\sqrt{2D\Delta t} \gtrsim h$; (iii) keep the lattice and select the leading eigenvalue among those with *positive real part*, discarding the parity mode explicitly. Whichever is chosen, state it.

3 1.5.2 — Stationary density

For the ergodic chain (no absorbing set; both boundaries reflecting), the stationary distribution solves

$$\rho_s^\top P^\pi = \rho_s^\top, \qquad \textstyle\sum_k \rho_s(k) = 1, \tag{8}$$

i.e. ρ_s is the left eigenvector of P^π with eigenvalue 1, equivalently the null vector of $(P^\pi - I)^\top$.

Solving the singular system. Do not use power iteration — it fails outright on the periodic lattice above. Instead replace one row of $A = (P^\pi - I)^\top$ by the normalization $\mathbf{1}^\top$ and solve the resulting nonsingular system $\tilde{A}\rho_s = e_N$ directly. This is one sparse solve and is robust to periodicity.

Marginalise and read off the potential. The position density is $\rho_s(i) = \sum_M \rho_s(i, M)$, and the operationally-defined effective potential of §1.3 follows with no fitting and no sampling noise:

$$U_{\text{meas}}(x_i) \;=\; -\ln \rho_s(i). \tag{9}$$

Key result

The chain settles the convention question outright. The Itô/Stratonovich/Hänggi ambiguity of §1.3 is a property of the *continuum limit*, not of the model. The chain *is* the model, so it has no ambiguity: it returns the true ρ_s exactly. The convention exponent α is then determined by regressing the chain's own $\ln \rho_s$ against $\ln D_{\text{eff}}$ (drift-corrected, per Estimator 1 of §1.3) with *zero* statistical error. What was a delicate measurement becomes a deterministic computation.

4 1.5.3 — Mean first-passage time

4.1 Derivation

Partition the states into the absorbing set $\mathcal{A}$ and the transient set $\mathcal{T}$. Let T_k be the expected number of steps to first enter $\mathcal{A}$ starting from k. Conditioning on the first step and using $T_l = 0$ for $l \in \mathcal{A}$:

$$T_k \;=\; 1 + \sum_l P_{kl} T_l \;=\; 1 + \sum_{l \in \mathcal{T}} P_{kl} T_l, \qquad k \in \mathcal{T}. \tag{10}$$

Collecting terms with $P^\pi_{\text{trans}} \equiv [P_{kl}]_{k,l \in \mathcal{T}}$ gives

Key result

$$\left(I - P^\pi_{\text{trans}}\right) T \;=\; \mathbf{1}, \qquad \langle T_{\text{esc}} \rangle = \Delta t \cdot T. \tag{11}$$

One sparse solve. Exact — no sampling error, no finite-horizon truncation, no unescaped trajectories to discard.

$I - P^\pi_{\text{trans}}$ is nonsingular precisely when every transient state can reach $\mathcal{A}$, which is check (S2). The factor Δt converts steps to physical time and is a common source of a silent factor-Δt discrepancy against Monte Carlo.

4.2 Which initial condition? (This is not a detail)

Equation (11) returns T_k for *every* starting state at once. Which entry to quote depends on what the Monte Carlo is measuring, and the two standard choices differ:

1. **Point start.** Trajectories released at x_1 with memory drawn from some p_0: quote $\sum_M p_0(M)\, T_{(i_1,M)}$. This includes the transient relaxation of the initial condition into the basin.

2. **Quasi-stationary start.** Escapes harvested from one long trajectory: the system has relaxed within the basin before escaping, so the correct object is the MFPT from the *quasi-stationary distribution* (QSD) ν, the normalized left eigenvector of P^{π}_{trans} for its leading eigenvalue λ_1. Escape from ν is exactly single-exponential and

$$\langle T_{\text{esc}} \rangle^{\text{QSD}} = \frac{\Delta t}{1-\lambda_1}. \tag{12}$$

Comparing a point-start chain number against a QSD-harvested Monte Carlo number (or vice versa) produces a real discrepancy that is nobody's bug. Decide which one the experiment is, and use it in all four methods.

Numerical check

Solver run on the Week 1 "medium" landscape ($\Delta U_{\text{trap}} = 0.5443$), $v = 1$, $\Delta t = 0.1$, $D = 0$, $h = v\Delta t$, $n = 99$, reflecting at $x_L = -5$, absorbing at $x_{\min} = 4.852$, 1-bit agent (7).

σ	$\Delta U_{\text{trap}}/\sigma$	exact chain	Monte Carlo ($N = 2 \times 10^4$)	ratio
0.35	1.555	1222.46	1219.29	0.9974
0.30	1.814	1413.48	1423.41	1.0070

Agreement to 0.3% and 0.7%, consistent with the $\sim 0.7\%$ standard error of an exponential mean at $N = 2 \times 10^4$. The chain construction and the simulator are mutually validated.

At $\sigma = 0.10$ ($\Delta U_{\text{trap}}/\sigma = 5.44$) the chain returns $\langle T_{\text{esc}} \rangle = 26678.5$ from a point start and $\Delta t/(1-\lambda_1) = 26645.4$ from the QSD — a 0.12% difference, confirming (12) and showing the transient correction is small when released at the basin bottom. Monte Carlo at this barrier would need $\sim 10^9$ steps for comparable precision; the chain takes milliseconds. *This is the entire argument for building it.*

5 1.5.4 — Committor

5.1 Derivation

Now use *two* absorbing sets: $\mathcal{A}$ (deep in the trap basin) and $\mathcal{B}$ (past the barrier). The committor q_k is the probability of reaching $\mathcal{B}$ before $\mathcal{A}$. With $q = 0$ on $\mathcal{A}$ and $q = 1$ on $\mathcal{B}$, conditioning on the first step gives for $k \in \mathcal{T}$

$$q_k = \sum_{l \in \mathcal{T}} P_{kl} q_l + \sum_{l \in \mathcal{B}} P_{kl}, \tag{13}$$

hence

Key result

$$\left(I - P^{\pi}_{\text{trans}}\right) q \;=\; P_{\mathcal{T}\to\mathcal{B}}\,\mathbf{1}. \tag{14}$$

The *same* matrix as (11), with a different right-hand side. Factorize $I - P^{\pi}_{\text{trans}}$ once and reuse it; the committor is then free.

5.2 What the committor buys

(i) The true transition state. The naive dividing surface is $x_{\min}$, but the surface that actually separates the basins is $\{q = \frac{1}{2}\}$. These need not coincide, and their separation is a physical statement about the barrier's shape.

(ii) The transmission factor κ, without fitting a memory kernel. Master-document item C5 notes that first passage to $x_{\min}$ measures a transition-state-theory time T_{TST} that underestimates the true residence time. Both times are now exact solves of (11) with different absorbing sets, so

$$\kappa \;=\; \frac{k_{\text{true}}}{k_{\text{TST}}} \;=\; \frac{T_{\text{TST}}}{T_{\text{true}}} \;\le\; 1 \tag{15}$$

is obtained *directly*, and $\kappa(m)$ is the Grote–Hynes transmission factor — measured rather than fitted, exactly as C5 recommends.

(iii) The memory diagnostic. The committor is a function of the *full* state, $q(i, M)$. This gives a sharp, quantitative test of whether memory is part of the reaction coordinate:

$$\boxed{\Delta q(i) \;\equiv\; \max_{M,M'} \left| q(i,M) - q(i,M') \right|} \tag{16}$$

If Δq is appreciable, then knowing the memory state changes the probability of escaping, and memory is mechanistically part of the transition — not merely a modifier of the drift. If $\Delta q \approx 0$, position alone is a sufficient reaction coordinate.

Numerical check

Committor computed for the 1-bit agent with $\mathcal{A}$ at $x = -1$, $\mathcal{B}$ at $x_{\min}$:

σ	$\max_i \Delta q(i)$	x at $q = \frac{1}{2}$
0.30	5×10^{-5}	3.30
0.10	1.7×10^{-4}	3.90
0.05	3.2×10^{-4}	4.10
0.02	8.4×10^{-4}	4.30
0.01	2.2×10^{-3}	4.40

Two results, both worth reporting.

The transition state is not at $x_{\min}$. It sits at $x \approx 3.3$–4.4, well *before* the concentration minimum at 4.85, and it migrates toward $x_{\min}$ as sensing improves. A trajectory that has reached $x_{\min}$ is already $\gtrsim 95\%$ committed; placing the absorbing boundary there therefore measures something close to the committed flux, and the recrossing correction κ is correspondingly mild. In the noisy regime the commitment happens a full 1.5 length units early.

Memory is not part of the reaction coordinate for this agent, at the 10^{-3} level across two decades of σ.

Key result

Why $\Delta q \approx 0$, and what it implies. This is not a null result — it follows from the structure derived in §1.1.2. Under independent readings, the two rows of the memory transition matrix are *identical*: the probability of arriving in the "up" state is $\Phi(s)$ regardless of where the agent came from. The bit is therefore re-randomized every step from the current position alone, and carries essentially no information forward in time.

So the 1-bit memory in this design does not *accumulate* evidence; it *converts* a temporal comparison into a drift, and that is all. The (x, M) chain nearly factorizes, escape is genuinely Markovian in x alone, and the clean Arrhenius law of §1.2 with a simple T_{eff} works precisely *because* of this.

This sharpens Q A.7 considerably. Non-exponential first-passage statistics should *not* be expected at $b = 1$ with independent readings, and looking for them there is looking in the wrong place. Genuine non-Markovian escape requires a design in which memory persists across steps — either the single-sensor variant (which the companion audit showed has real action autocorrelation, $\langle a_t a_{t+2} \rangle = -\frac{1}{3}$ exactly), or a $b \geq 2$ accumulator that integrates evidence over several steps. *That* is where the project should look, and $\Delta q(i)$ is the instrument that will detect it.

6 1.5.5 — Full first-passage-time distribution from the spectrum

6.1 Survival probability in closed form

With initial distribution p_0 on $\mathcal{T}$, survival to step t requires never having entered $\mathcal{A}$:

$$S(t) \;=\; \Pr(T_{\text{esc}} > t) \;=\; p_0^{\top} \left(P^{\pi}_{\text{trans}}\right)^{t} \mathbf{1}, \qquad f(t) \;=\; \Pr(T_{\text{esc}} = t) \;=\; S(t-1) - S(t). \tag{17}$$

Since P^{π}_{trans} is sub-stochastic and irreducible-to-$\mathcal{A}$, all its eigenvalues satisfy $|\lambda| < 1$. Diagonalising $P^{\pi}_{\text{trans}} = \sum_k \lambda_k u_k w_k^{\top}$,

Key result

$$S(t) \;=\; \sum_k c_k\, \lambda_k^{\,t}, \qquad c_k = \big(p_0^{\top} u_k\big)\big(w_k^{\top}\mathbf{1}\big). \tag{18}$$

The first-passage distribution is a *single exponential if and only if one term contributes.* Departure from exponentiality is therefore read directly off the spectrum — no histogram, no fitting window, no binning choice.

As a consistency check, summing (17) over $t \geq 0$ gives $\sum_t S(t) = p_0^{\top}(I - P^{\pi}_{\text{trans}})^{-1}\mathbf{1}$, which is exactly the mean from (11). The two routes agree identically, which is a useful unit test.

6.2 The two timescales

$$\tau_{\text{esc}} = -\frac{\Delta t}{\ln \lambda_1} \;\approx\; \frac{\Delta t}{1-\lambda_1} \qquad \text{(escape; the slow mode)}, \tag{19}$$

$$\tau_{\text{rel}} = -\frac{\Delta t}{\ln |\lambda_2|} \qquad \text{(intra-basin relaxation; the fast mode)}. \tag{20}$$

Clean Poissonian escape requires $\tau_{\text{rel}} \ll \tau_{\text{esc}}$. The dimensionless non-exponentiality measure is $|\lambda_2|/\lambda_1$, or equivalently $\tau_{\text{rel}}/\tau_{\text{esc}} = \ln \lambda_1 / \ln |\lambda_2|$.

Key result

The chain hands you the Week 3 fitting window. The Week 3 instruction "fit only the tail, $t \gg \tau_{\text{rel}}$" is unquantified as stated, and a student has no principled way to pick the cut. The spectrum supplies it: $\tau_{\text{rel}} = -\Delta t / \ln |\lambda_2|$, computed exactly, before any histogram is made. Report the window as a number, not a judgement call.

Numerical check

At $\sigma = 0.10$ the transient spectrum is

$$\lambda \;\in\; \{\, 0.999996,\;\; 0.997428,\;\; -0.993404,\;\; -0.999993,\;\; \ldots \,\}.$$

Reading off the physical modes: $\tau_{\text{esc}} = -\Delta t/\ln(0.999996) \approx 2.66 \times 10^4$ and $\tau_{\text{rel}} = -\Delta t/\ln(0.997428) \approx 38.9$. The separation is $\tau_{\text{rel}}/\tau_{\text{esc}} \approx 1.5 \times 10^{-3}$: escape is single-exponential to high accuracy, as anticipated in §5. **The Week 3 histogram should be fitted only for $t \gg 39$ time units.**

And note the artifact. The second-largest eigenvalue *by magnitude* is -0.999993, not 0.997428. That mode is the lattice parity discussed in §2, and taking it would report $|\lambda_2|/\lambda_1 \approx 1$ — i.e. "severe non-exponentiality" — for a chain that is in fact cleanly exponential. Select the physical relaxation mode explicitly, or remove the periodicity.

7 1.6 — The four-way cross-validation

7.1 1.6.1 — The four routes and what each assumes

Route	Computes	Assumes beyond the model
Monte Carlo	$\langle T_{\text{esc}} \rangle$ by direct simulation	nothing; error is statistical, $\propto N^{-1/2}$
Exact chain	$\langle T_{\text{esc}} \rangle$ by (11)	only the position discretization — and *nothing at all* if $h = v\Delta t$, $D = 0$
MFPT quadrature	$\langle T_{\text{esc}} \rangle$ by the §1.4 double integral	the continuum diffusion limit: one-step moments, a convention α, no action autocorrelation
Saddle-point Kramers	$\langle T_{\text{esc}} \rangle$ by the asymptotic formula	all of the above, *plus* $\Delta U_{\text{eff}}/T_{\text{eff}} \gg 1$

The four form a *ladder of increasingly strong assumptions*. That structure is what makes disagreements diagnostic: each rung adds exactly one assumption, so a disagreement localises to the rung where it appears.

7.2 1.6.2 — What each pairwise disagreement means

Pair	A disagreement means ...
MC ↔ chain	These describe the *same* discrete-time model and must agree to sampling error. A gap is a **bug**, a grid mismatch ($h \neq v\Delta t/K$), a boundary-set mismatch, an initial-condition mismatch (§4.2), or a missing factor Δt. This is the primary gate; fix it before looking at anything else.
chain ↔ quadrature	The *continuum limit is failing*. Three candidates, in order of likelihood: (i) the convention exponent $\alpha \neq 1$ and the correction $\mu_{\text{Itô}} = \mu + (1-\alpha)D'_{\text{eff}}$ has not been applied; (ii) action autocorrelation invalidates the one-step D_{eff}, so the Green–Kubo-corrected D^{∞}_{eff} is needed; (iii) $v\Delta t \not\ll w_1$, breaking assumption (A1).
quadrature ↔ saddle	The barrier is not asymptotically deep, condition (V3). At $\Delta U_{\text{eff}}/T_{\text{eff}} \sim 2$–$6$ a discrepancy of tens of percent is **expected and quantitative**, not a defect. Report it as the size of the prefactor correction; never fit a measured slope against the asymptotic one in this regime.
MC ↔ saddle	Compounds every assumption at once and localises nothing. Useful only as a final headline number after the other three pairs agree.

Key result

The four-way test would have caught the audit finding automatically. The companion document showed that the single-sensor 1-bit agent has $\langle a_t a_{t+2} \rangle = -\frac{1}{3}$, which suppresses the long-time diffusivity by $\kappa_D \approx 0.523$ relative to the one-step formula. The chain (which tracks the true dynamics) and the quadrature (which uses the one-step D_{eff}) would then disagree by a factor of roughly $1/\kappa_D$ — a large, obvious, correctly-localised failure on the chain↔quadrature rung. This is precisely what the cross-validation is *for*: it converts a subtle modelling error into a loud numerical one.

7.3 Matching conventions across the four methods

Most spurious disagreements come from comparing subtly different quantities. Fix all five of these once, in a shared configuration object, and use them everywhere:

(M1) **Absorbing set.** The same set of positions in all four. If the chain absorbs at bin $n-1$ and the quadrature at $b = x_{\min} + \delta$, they are different problems.

(M2) **Reflecting boundary.** The quadrature needs an explicit r; the chain and MC must clamp at the same place, and it must be far enough left that the answer is insensitive to it (check this).

(M3) **Initial condition.** Point start versus QSD, per §4.2. Chain and MC must use the same; the quadrature's $T(x)$ is a point start by construction.

(M4) **Time units.** The chain and MC count *steps*; the quadrature and Kramers return *time*. Multiply by Δt exactly once.

(M5) **Which D_{eff}.** If action autocorrelation is present, the quadrature must use D_{eff}^{∞}, not the one-step D_{eff}.

7.4 Order of operations

1. Build the chain first. It is the cheapest and the most nearly exact.
2. Validate the Monte Carlo against it at *shallow* barriers, where MC is affordable. This is the real purpose of the shallow settings from Week 1.
3. Only then compare the quadrature, and interpret any gap using the middle row of the table above.
4. Add the saddle-point number last, and use the disagreement to *locate* the asymptotic regime rather than to test the theory.
5. Report all four in a single table, at matched ΔU_{trap}. That table is the end-of-month deliverable.

8 Summary and implementation checklist

Object	Computation	Eq.
Transition matrix	$P^{\pi}_{(i,M)\to(j,M')} = \sum_a \pi(a\|i,M) K_a(i,j) q(M'\|M,i,j,a)$	(4)
1-bit chain	two nonzeros/row, $p_{\text{persist}} = \Phi\big(\Delta c/\sqrt{2}\sigma\big)$	(7)
Stationary density	null vector of $(P^{\pi} - I)^{\top}$, row-replaced	—
MFPT	$(I - P^{\pi}_{\text{trans}})T = \mathbf{1}$	(11)
QSD escape time	$\Delta t/(1-\lambda_1)$	(12)
Committor	$(I - P^{\pi}_{\text{trans}})q = P_{\mathcal{T}\to\mathcal{B}}\mathbf{1}$, same matrix	(14)
Survival	$S(t) = p_0^{\top}\big(P^{\pi}_{\text{trans}}\big)^t\mathbf{1} = \sum_k c_k\lambda_k^t$	(18)
Fitting window	$\tau_{\text{rel}} = -\Delta t/\ln\|\lambda_2\|$ (physical mode)	—

Checklist

(C1) Grid matched, $h = v\Delta t/K$; at $D = 0, K = 1$ state explicitly that the chain is exact.

(C2) Row sums $= 1$ to machine precision; every transient state reaches $\mathcal{A}$.

(C3) Periodicity resolved (rest action, $D > 0$, or explicit mode selection) — and the choice stated.

(C4) Initial condition declared: point start or QSD, used consistently in all four methods.

(C5) MFPT and committor share one factorization of $I - P^{\pi}_{\text{trans}}$.

(C6) Physical λ_2 selected, not the parity mode; τ_{rel} reported as the Week 3 fitting window.

(C7) $\Delta q(i)$ reported per memory class — it is the sharpest available test of whether memory is mechanistically involved.

(C8) Four-way table assembled at matched ΔU_{trap}, with disagreements localised by rung rather than averaged away.